TABLE I
NETWORK DATA

Transmission line (n,m)	1-3	3-9	6-10	10-11	11-14	14-16
$\overline{S}_{nm}$ [p.u.]	2.3	1.6	2.5	3.0	2.3	2.1

TABLE II
DATA FOR GENERATING UNITS

Unit (i)	node	$\underline{P}_i$ [p.u.]	$\overline{P}_i$ [p.u.]	$\underline{Q}_i$ [p.u.]	$\overline{Q}_i$ [p.u.]	R_i^{U}, R_i^{D} [p.u.]	R_i^{+}, R_i^{-} [p.u.]	C_i $[\frac{\$}{\mathrm{p.u.}}]$	λ_i^{SU} [$]	u_i^{ini}	P_i^{ini} [p.u.]	r_i^{ini} [p.u.]
1,2	1	0.100	0.200	0.00	0.10	0.000	0.100	1109	300	1	0.20	0.00
3,4	1	0.152	0.760	-0.25	0.30	0.100	0.500	1246	400	1	0.76	0.00
5,6	2	0.100	0.200	0.00	0.10	0.000	0.100	1109	300	1	0.20	0.00
7,8	2	0.152	0.760	-0.25	0.30	0.100	0.500	1246	400	1	0.76	0.00
9	7	0.800	3.500	-0.25	1.50	1.000	2.500	1720	100	0	0.00	0.00
10,11	7	0.150	1.000	0.00	0.60	0.550	0.850	1660	275	1	0.55	0.45
12-14	13	0.620	1.970	0.00	0.80	0.450	1.150	1408	300	1	1.97	0.00
15-19	15	0.024	0.120	0.00	0.06	0.096	0.096	2141	400	0	0.00	0.00
20	15	0.500	1.550	-0.50	0.80	0.450	1.000	1592	200	1	1.10	0.45
21	16	0.500	1.550	-0.50	0.80	0.450	1.000	1592	200	1	1.10	0.45
22	18	1.000	4.000	-0.50	2.00	1.500	2.800	1917	250	0	0.00	0.00
23	21	1.000	4.000	-0.50	2.00	1.500	2.800	1917	250	0	0.00	0.00
24-29	22	0.000	0.500	-0.10	0.16	0.150	0.500	0	100	1	0.50	0.00
30,31	23	0.500	1.550	-0.50	0.80	0.450	1.000	1592	200	1	1.10	0.45
32	23	0.800	3.500	-0.25	1.50	1.000	2.500	1720	100	0	0.00	0.00

TABLE III
LOAD FACTOR CORRESPONDING TO EACH TIME PERIOD

Time period	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8	t_9	t_{10}	t_{11}	t_{12}
Load factor	0.75	0.70	0.65	0.60	0.62	0.63	0.65	0.68	0.70	0.72	0.75	0.78
Time period	t_{13}	t_{14}	t_{15}	t_{16}	t_{17}	t_{18}	t_{19}	t_{20}	t_{21}	t_{22}	t_{23}	t_{24}
Load factor	0.80	0.85	0.85	0.90	0.92	0.95	0.98	1.00	0.97	0.93	0.91	0.92

IV. CASE STUDY

This section presents numerical results for a case study based on the IEEE one-area 24-node reliability test system (RTS) [24]. In this study, a daily time horizon (time periods t_1 to t_{24}) is considered. Under the per-unit system used, 1 p.u. of power is equivalent to 100 MW.

A. Data

Network data $\overline{S}_{nm}$ are those reported in [24], except the ones given in Table I. In addition, data for generating units are given in Table II. Note that generating units 1, 2, 5 and 6 do not provide reserve. Note also that to highlight voltage issues, the synchronous condenser connected to node 14 in [24] is removed.

Hourly active and reactive loads are those given in [24] multiplied by the hourly load factors provided in Table III. Accordingly, time period t_{20} is the peak hour.

Two wind farms located at nodes 3 and 14 are considered, whose installed capacities are 2.85 p.u. and 2.96 p.u., respectively. Total wind capacity (i.e., 5.81 p.u.) is 13.71% of the total installed capacity (42.36 p.u.). The wind power production uncertainties are characterized through 40 scenarios, as illustrated in Fig. 3, whose upper plot corresponds to the farm located at node 3, and whose lower plot refers to the farm located at node 14. Note that each dot illustrates the wind production level of the corresponding farm for a given time period under a particular scenario. In other words, each scenario of any of the two plots contains 24 dots indicating the wind production levels of the corresponding farm over 24 time periods. These 40 scenarios have been derived from historical data, and accurately represent the uncertainty without making the considered problem computationally intractable. The corresponding probabilities (ρ_s) are given in Table IV. Based on the scenarios considered, the expected total wind power production over the 24 time periods is 1.72 p.u. (i.e., 29.60% of the total wind capacity).

The value of load shedding for all demands (V_d^{SH}) is considered 10 000 $/p.u. Finally, tolerance ε required for convergence check is set to 0.3% of the value of the objective function (expected cost).

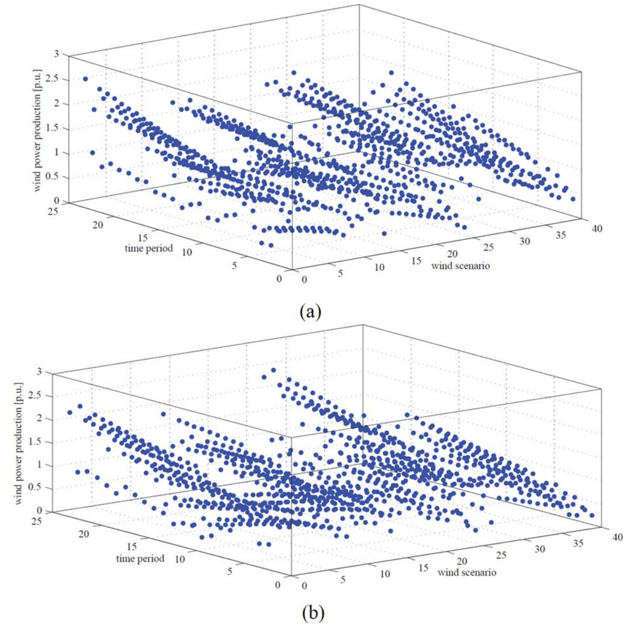

Fig. 3. Wind power scenarios of the farm (a) located at node 3, (b) and of that located at node 14.

TABLE IV
PROBABILITY OF EACH SCENARIO (ρ_s)

s	ρ_s	s	ρ_s	s	ρ_s	s	ρ_s	s	ρ_s	s	ρ_s	s	ρ_s	s	ρ_s
s_1	0.01	s_6	0.03	s_{11}	0.05	s_{16}	0.01	s_{21}	0.03	s_{26}	0.01	s_{31}	0.02	s_{36}	0.03
s_2	0.01	s_7	0.03	s_{12}	0.05	s_{17}	0.01	s_{22}	0.04	s_{27}	0.01	s_{32}	0.02	s_{37}	0.03
s_3	0.01	s_8	0.03	s_{13}	0.01	s_{18}	0.02	s_{23}	0.04	s_{28}	0.01	s_{33}	0.02	s_{38}	0.04
s_4	0.02	s_9	0.04	s_{14}	0.01	s_{19}	0.02	s_{24}	0.05	s_{29}	0.01	s_{34}	0.02	s_{39}	0.04
s_5	0.02	s_{10}	0.04	s_{15}	0.01	s_{20}	0.03	s_{25}	0.05	s_{30}	0.01	s_{35}	0.02	s_{40}	0.04

B. Cases Considered

Three different cases are analyzed:

Case A) This case refers to a dc-UC problem, where both stages embody a dc representation of the network. This problem is directly solved using a mixed-integer linear solver.